

Linear Algebra - Homework 1

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Section A

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Attempted Solution:

251-3207 

a) System of Three Linear Equations:-

CPU: $2x_1 + 4x_2 + 6x_3 = 40$

Memory: $x_1 + 3x_2 + 4x_3 = 25$

Storage: $x_1 + 2x_2 + 5x_3 = 30$

b) Matrix Form of the System of Equations:-

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 4 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 40 \\ 25 \\ 30 \end{bmatrix}$$
$$A \cdot x = b$$

c) Augmented Matrix:-

$$\left[\begin{array}{ccc|c} 2 & 4 & 6 & 40 \\ 1 & 3 & 4 & 25 \\ 1 & 2 & 5 & 30 \end{array} \right]$$

d) Conversion to Row Echelon Form:-

$$R_1' = \frac{1}{2} R_1$$

$$R_2' = R_2 - R_1'; R_3' = R_3 - R_1'$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 20 \\ 1 & 3 & 4 & 25 \\ 1 & 2 & 5 & 30 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 20 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 2 & 10 \end{array} \right]$$

$$R_3' = \frac{1}{2}R_3$$

$$\begin{bmatrix} 1 & 2 & 3 & ; & 20 \\ 0 & 1 & 1 & ; & 5 \\ 0 & 0 & 1 & ; & 5 \end{bmatrix}$$

$\therefore$ Hence, REF is found.

e) Values of Variables.

Let Type A VM = x_1 ,

Type B VM = x_2

& Type C VM = x_3 ; From REF, we found that

$$\boxed{x_3 = 5}$$

Put $x_3 = 5$ in R_2

$$0 + (x_2) + (x_3) = 5$$

$$5 + x_2 = 5$$

$$\boxed{x_2 = 0}$$

Put $x_2 = 0$ & $x_3 = 5$ in R_1

$$x_1 + 2(0) + 3(5) = 20$$

$$x_1 = 20 - 15$$

$$\boxed{x_1 = 5}$$

$\therefore$ Hence values of VMs are

Type A = 5

Type B = 0

Type C = 5

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f) Verification

$$2x_1 + 4x_2 + 6x_3 = 40 \dots \textcircled{i}$$

$$x_1 + 3x_2 + 4x_3 = 25 \dots \textcircled{ii}$$

$$x_1 + 2x_2 + 5x_3 = 30 \dots \textcircled{iii}$$

Put values of x_1, x_2 & x_3 in each eq, one by one

$$\bullet \quad 2(5) + 0 + 6(5) = 40$$

$$10 + 30 = 40$$

$$40 = 40 \text{ eq } \textcircled{i} \text{ verified}$$

$$\bullet \quad (5) + 0 + 4(5) = 25$$

$$5 + 20 = 25$$

$$25 = 25 \text{ eq } \textcircled{ii} \text{ verified}$$

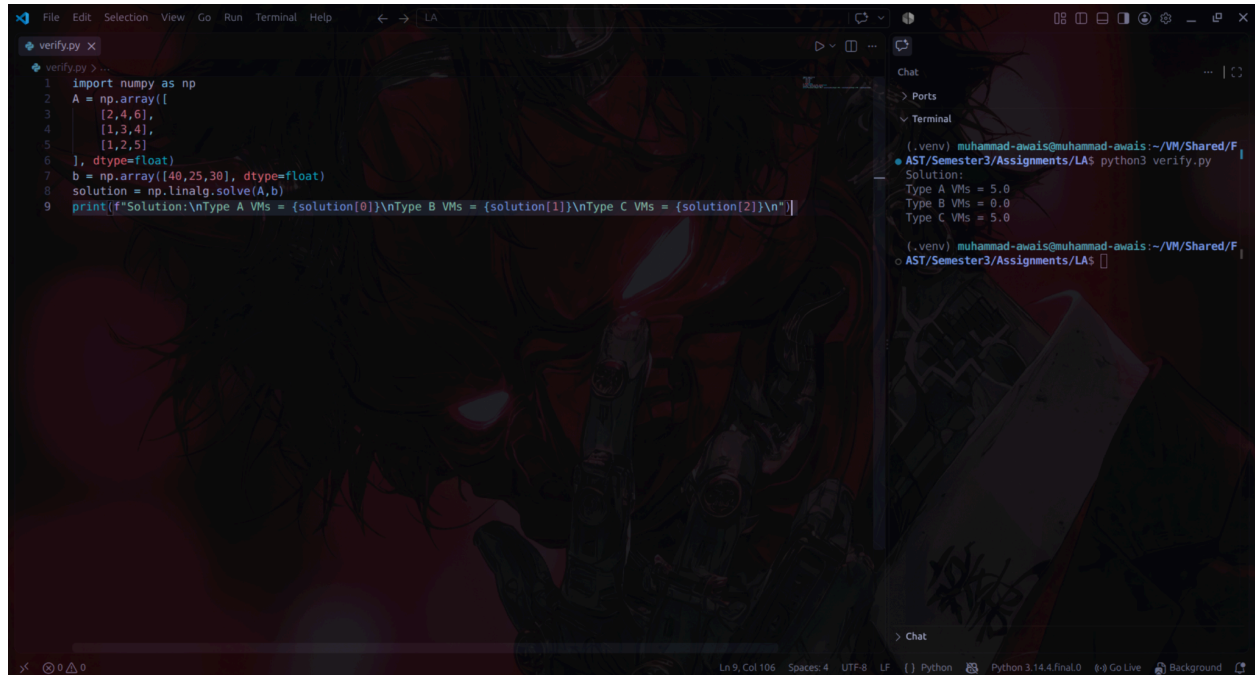
$$\bullet \quad (5) + 0 + 5(5) = 30$$

$$5 + 25 = 30$$

$$30 = 30 \text{ eq } \textcircled{iii} \text{ verified}$$

Type A VMs	5
Type B VMs	0
Type C VMs	5

Python Verification:



```
1 import numpy as np
2 A = np.array([
3     [2,4,6],
4     [1,3,4],
5     [1,2,5]
6 ], dtype=float)
7 b = np.array([48,25,30], dtype=float)
8 solution = np.linalg.solve(A,b)
9 print(f'Solution:\nType A VMs = {solution[0]}\nType B VMs = {solution[1]}\nType C VMs = {solution[2]}\n')
```

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Ports

Terminal

```
(.venv) muhammad-awais@muhammad-awais:~/VM/Shared/F
● AST/Semester3/Assignments/LAs python3 verify.py
Solution:
Type A VMs = 5.0
Type B VMs = 0.0
Type C VMs = 5.0

(.venv) muhammad-awais@muhammad-awais:~/VM/Shared/F
○ AST/Semester3/Assignments/LAs
```

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